



Learning Mobility for Real Legged Robots

Fact Sheet

Project Information

LeMo

Grant agreement ID: 852044

[Project website](#) **DOI** [10.3030/852044](https://doi.org/10.3030/852044)

Project closed

EC signature date

8 October 2019

Start date

1 January 2020

End date

31 December 2024

Funded underEXCELLENT SCIENCE - European
Research Council (ERC)**Total cost**

€ 1 496 370,00

**EU
contribution**

€ 1 496 370,00

Coordinated byEIDGENOESSISCHE
TECHNISCHE
HOCHSCHULE ZUERICH
Switzerland

Project description

Robots get a leg up on locomotion

The robotics industry has been an important cornerstone of modern automated industrial processes. It's also one of the most heavily researched topics for applications in commercial and industrial sectors. Working leg mechanisms have recently been heavily researched as they provide robots with increased mobility and the ability to adapt to most terrain. Yet these advances still face limits when it comes to complex situations. The EU-funded LeMo project plans to research methods by

which learned behaviours can be directly transferred from simulations to the robots to provide them with more advanced motion capabilities. This way they aim to surpass current limitations, providing a better understanding of robot locomotion.

Objective

Research and applications in legged robotics has made significant progress over the last decade, driven by more advanced actuation systems, better on-board computation, and significantly improved sensors for perceiving the environment. State-of-the-art model-based planning and control algorithms can plan for contact points and body motions to move legged systems over complex environments. However, these methods have shown clear performance limits when it comes to behaviours and situations that are more complex, and it is unclear if and how these limits can be overcome with classical control methods. On the other hand, recent advances in reinforcement learning has put forward unprecedented capabilities to learn control policies for complex behaviours.

With our preliminary findings, we were the first group to present methods that allow directly transferring learned behaviours from simulation to reality to create advanced motion skills for complex legged robots. This breakthrough has the potential to revolutionize the field of legged locomotion control. In this ERC project, we want to research this highly promising area and investigate the use of machine learning tools to make legged robots autonomously move in realistic outdoor scenarios. In three parallel research streams, we will learn how to accurately model the system dynamics from experience, how to abstractify, generate and coordinate different complex behaviours that involve multi-contact situations, and how to combine these with perception to enable autonomous navigation in complex environments. The proposed methods have the potential to overcome the limitations of commonly used optimization-based methods such as limited model accuracy, local minima, conservative performance, computational load and execution time, and it will help us to better understand the fundamentals of locomotion in robots and biology.

Fields of science (EuroSciVoc)

[engineering and technology](#) > [electrical engineering](#), [electronic engineering](#), [information engineering](#) > [electronic engineering](#) > **[sensors](#)**

[engineering and technology](#) > [electrical engineering](#), [electronic engineering](#), [information engineering](#) > [electronic engineering](#) > **[robotics](#)**



Keywords

[Legged Robotics](#)

[Locomotion Control](#)

[Learning Locomotion](#)

[Machine Learning](#)

Programme(s)

H2020-EU.1.1. - EXCELLENT SCIENCE - European Research Council (ERC) MAIN PROGRAMME

[See all projects funded under this programme](#)

Topic(s)

[ERC-2019-STG - ERC Starting Grant](#) 

[See all projects funded under this topic](#)

Funding Scheme

ERC-STG - Starting Grant

[See all projects funded under this funding scheme](#)

Call for proposal

[ERC-2019-STG](#) 

[See all projects funded under this call](#)

Host institution



EIDGENOESSISCHE TECHNISCHE HOCHSCHULE ZUERICH

Net EU contribution 

€ 1 496 370,00

Total cost 

€ 1 496 370,00

Address

Raemistrasse 101

8092 Zuerich

Switzerland 

Region

Schweiz/Suisse/Svizzera > Zürich > Zürich

Activity type

Higher or Secondary Education Establishments

Links

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[Participation in EU R&I programmes](#) 

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Beneficiaries (1)



EIDGENOESSISCHE TECHNISCHE HOCHSCHULE ZUERICH

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Last update: 20 October 2025

Permalink: <https://cordis.europa.eu/project/id/852044>

European Union, 2026